

MGTA 456 Supply Chain Analytics

Winter 2026

Prof. Robert Montgomery

Email: romontgomery@ucsd.edu

Office Hours: by appointment

Teaching Assistant: Raunak Sengupta

Email: r2sengupta@ucsd.edu

Course Location: OTRSN 1E107, Mondays, 11:00am-1:50pm.

DESCRIPTION

This course is designed to link theory with application to use data analytics to manage modern supply chains. The course will establish which components constitute a supply chain and how to understand the strategies to improve their operations. We will explore both classical along with cutting edge approaches to supply chain management. As much of supply chain management deals with uncertainty both in terms of demand for products and supply-side uncertainty from disruptions and/or delays, we will focus on how to use data to better account for uncertainty. We will spend time on current areas of focus within the supply chain literature, especially on the challenges of last-mile delivery, and how certain companies and researchers are working to overcome these challenges.

OBJECTIVES

At the close of this course, you will be able to:

- Map the key components of a company's supply chain
- Identify the challenges and sources of uncertainty within a supply chain
- Use data to analyze a company's supply chain and effectively communicate improvement strategies
- Discuss approaches companies are using to modernize supply chains
- Improve your capabilities with Python and Excel

COURSE MATERIALS

Required

- Supply Chain Management: Strategy, Planning, and Operation 7th Edition, by Sunil Chopra.

SCHEDULE

Date	Class Topic	Class Prep work
Week 1 January 5	How to Approach Analyzing a Supply Chain & Newsvendor Models	-Textbook Chapters 1, 2, and 13
Week 2 January 12	Demand Forecasting	-Textbook Chapters 3 and 7
Week 3 January 19 (no class)	Moving Averages and Exponential Smoothing	-Textbook Chapter 4
Week 4 January 26	Supply Chain Networks	-Textbook Chapter 11

Week 5 February 2	Cycle Inventory	-Textbook Chapter 5
Week 6 February 9	Inventory Control	-Textbook Chapters 10 and 12
Week 7 February 16 (no class)	Survey of New Approaches to Supply Chains	Reading prep material to be posted on canvas
Week 8 February 23	Globalization and Sustainability	Reading prep material to be posted on canvas
Week 9 March 2	Last Mile Delivery	Reading prep material to be posted on canvas
Week 10 March 9	Overview of Revenue Management	Final Project Due by the start of class

For the classes which fall on holidays (weeks 3 and 7) the professor will prerecord lectures covering the material.

ASSESSMENTS

In lieu of exams, this course will assess student progress through four primary means: Case-write up, Quizzes, Homework Assignments, and a Final Project.

Case Writeup

There will be a single case writeup which will be completed in groups. More details on expectations for this assignment will be provided in class.

Homework Assignments

There will be between 5 and 7 homework assignments scheduled throughout the quarter. Students are allowed to collaborate on these homework assignments, but must turn in their own reports and code when requested.

Quizzes

This class will include regular quizzes, which will occur at the start of class to ensure students have reviewed the material prior to class. These quizzes will also allow the professor to understand which topics require additional attention. The lowest quiz grade for each student will be dropped. This will allow students to miss one class without penalty. The professor will post an overview before each quiz, addressing which material is fair game for that week's quiz.

Final Project

There will be a final project to be completed in groups, and due before the start of the last class. Guidelines for the final project will be posted on Canvas.

Final Exam

There will not be a final exam for this class.

GRADING

Assignments	Percentage
Class Participation	10%
Case Write-Ups	10%
Homework Assignments	25%
Quizzes	25%
Final Project	30%
Total	100%

Large Language Models AND OTHER AI TOOLS

LLMs and other AI tools are now widely available. In your careers you will have to figure out when these tools are helpful and when they are not. I encourage you to use AI tools in any way that will help guide your learning. I am also learning about the capabilities of these tools. So, for the benefit of myself and the class, please let me know on assignments if you discovered a particularly useful application for these tools.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.